

Coding Assessment Preparation

The Hackerrank website has challenges you can work through to practice your coding skills. There is a dynamic programming section containing problems that you can work on in timed conditions.

<https://www.hackerrank.com/domains/algorithms/dynamic-programming/page/1>

Topics to Revise

- Data Structures and Algorithms
- Mathematics and Statistics
- Conditional Probability
- Database Management
- General Aptitude and Logical Reasoning

Data Structures and Algorithms

1. Given an array of integers, find the maximum product of two integers in the array.
2. Write a function to reverse a singly linked list.
3. Given a string containing only (,) , { , } , [, and] , write a function to determine if the string is valid. A valid string requires that all brackets are closed in the correct order.
4. Given a sorted array of integers and a target value, implement binary search to return the index of the target. If it is not found, return -1.
 - a. What is the time complexity of your solution?
 - b. How would you modify it to return the insertion index instead?
5. Implement an in-order traversal of a binary tree and return the list of visited nodes.
6. Implement the quicksort algorithm to sort an array of integers.
7. Write a function to calculate the nth Fibonacci number using dynamic programming
8. Given a sorted array of integers, write a function to determine the index of a given target value. If the target does not exist in the array, return -1.
9. Given a set of coin denominations and a target amount, determine the minimum number of coins needed to make that amount. Assume an infinite supply of each coin denomination.

Mathematics, Probability and Statistics

1. A deck of 52 playing cards is shuffled, and you draw two cards at random. What is the probability that both cards are aces?
2. Find the greatest common divisor (GCD) of 56 and 98 using the Euclidean algorithm
3. Given two vectors $v = [3, 4]$ and $w = [1, 2]$, calculate the dot product of v and w .
4. A stock has a 60% probability of increasing in value on any given day. If it increases today, the probability that it increases tomorrow is 70%. If it does not increase today, the probability that it increases tomorrow is 40%. What is the probability that the stock increases over two consecutive days?
5. A test for a rare disease has a 95% accuracy rate for detecting the disease when it is present (true positive rate) and a 90% accuracy rate for correctly identifying when it is absent (true negative rate). If 1 in 1000 people have the disease, what is the probability that a person who tests positive actually has the disease?

Arrays

1. Given an array of integers, write a function to find the contiguous subarray with the maximum sum. Return the sum of this subarray.
2. Write a function that rotates an array to the right by k steps, where k is a non-negative integer.
3. Given an integer array `nums` of length n where each element is between 1 and n , inclusive, find all the duplicate numbers in the array. Assume that each number appears at most twice.

ADDITIONAL QUESTIONS

1. You have a stack of coins arranged in rows. You can add water drops, where each drop stacks on top of the coins until it overflows. For example, if a column of coins is two units high, you can add up to 2 water drops to that column. Write a program to calculate the total number of water drops you can add before any column overflows.
2. Write code to generate the nth Fibonacci number. What is the time complexity of your solution?
3. Given an array of numbers, find the pairs of numbers that add up to a given target number k.
4. You are given an array formatted like [1, 4, -8, 4]. Find the biggest contiguous subarray value. For example two possible subarrays are [1, 4] or [-8, 4]. The one that summed up and has the biggest value is the first. This needs to be done with all possible subarrays. What is the time complexity?
5. Given an array with 1 number that is not repeated more than once, find the number not repeated.
In a list of integers, find the biggest sub list with increasing numbers. The order shall stay the same but you can skip some numbers. Example: [1,2,2,3] shall return [1,2,3]; [1,2,6,3,2,4] shall return [1,2,3,4]
6. A person wants to watch movies. Each movie has a length from 1.01 time unit to 3 time units. The person can watch for 3 time units per day. What is the least amount of days he needs to watch all of the movies?
7. Find max size of sub-squares in a matrix (dynamic programming)
8. Contiguous subarray of maximum sum within an array (give indices of the array and the maximum sum).
 - a. Given an array, how to design a program finding a subarray whose sum equals to 0.
 - b. Hypothesis of linear regression.
 - c. Linear regression
 - d. To draw randomly words from a file with constraints being added.
 - e. Complexity of accessing an element within an array
9. Have a string of parameters with open brackets. You are required to ensure all brackets are closed by inserting additional brackets. Have to calculate the number of brackets to close all parentheses.
10. Given a list of integers and a parameter k, where k represents the number of times an operation can be applied to a single element, apply a sequence of operations to minimize the total sum of the list. Write a program to determine the minimum possible sum after applying the operations k times in total
11. Given an array, code an algorithm to find a (contiguous) sub-array which has the maximum sum
 - a. What is overfitting?
 - b. What is Monte Carlo?
 - c. Discussion about statistical learning methods, their benefits, cons etc.
 - d. Questions about black Scholes and option pricing.
 - e. Questions about call/put options and hedged portfolio.
 - f. Discussion about bootstrapping and swap curve building
12. Partition a list in 2 sublists, the first should be shorter and of greater sum than the second?
13. You have n stairs. You can take either 1, 2 or 3 steps. How many ways are there to climb the stairs?
14. You have two arrays A, B of length n. An action is to take two indices $0 < i, j < n$, swap the values in A[i] and B[j]. After k such actions. How many unique numbers are present in A?
15. Write a function max_average(a) that takes an array of integers 'a' as an input and output the largest average of any subarray of 'a'
 - A subarray is any array consisting of consecutive elements of 'a'. E.g. if a = [a1, a2, a3, a4], then [a2, a3, a4] is a subarray of length 3 but [a1, a3, a4] is not a subarray.
16. You are given an integer array 'arr' of length 'n' e.g. [1, 5, 6, 7]. The allowed operation is:
 - choose an array position i that is not the first
 - from that position, choose integer x where $1 \leq x \leq \text{arr}[i]$
 - subtract x from arr[i]
 - add x to arr[i-1]
 - After any number of these operations, minimise max(arr).
 - What is the time complexity of your solution?
18. If you have stacks and want to create a data queue, how would you do it?
19. Develop a function for a moving average, given an array of 5-10 numbers and a window size.
20. Given an array of integers and a target value, write a function to find and return two distinct numbers from the array whose sum equals the target. Return the pair as a subarray. Assume exactly one solution exists
21. Integer match
arr1 = [123, 456]
arr2 = [321, 183]

Each step, you can increment or decrement one digit of a number in either arr, for example 123 --> 223 or 133 or 122

Return the minimal total steps to make arr1 == arr2

22. min_cost

The cost of an arr is defined by $\sum((arr[i]-arr[i-1])**2)$, eg. cost of [1,2,4] is $(2-1)**2 + (4-2)**2 = 5$

You can insert any number at any place of the arr, return the possible minimal cost after insertion

23. Can do two operations to a tuple (a,b) --> (a+b, b) or (a, a+b)

isPossible(a,b,c,d) return "Yes" if after a series of operations to (a,b), it can equal to (c,d)
otherwise return "No"

24. Discuss Python's features, object oriented programming, decorator.

25. Given 3 integers: d, t and n. d is the number of devs, t is the number of testers and n is the length of a sequence. Have to build a sequence/array of programmers (d and t) to outline a team, i.e. ddt would be a team of 2 developers (d=2) and 2 testers (t=2). Calculate the number of possible sequences when d, t and n are specified.

26. Optimization question: Given an array of ints, find the highest integer for each cell such that: it is lower or equal to the cell, no two numbers are the same, and their sum is the maximal possible.

27. Given two integers, determine the side length of the largest square you can form using units from both numbers, where each square requires 4 equal units. You may not transfer leftover units from one number to the other. Provide an O(1) time solution.

28. Graph question: Given a tree (not necessarily binary), find the number of nodes that have balanced branches (meaning the number of descendants are equal in each branch).

29. We have a list A of values of different sizes (different bars of chocolate) and k (the number of children). Find an efficient algo to find the maximal size s.t. the bars of chocolate can be divided and equally given to each of the children.

30. Find an efficient algorithm that outputs the number of pairs of a list A s.t. $A[i] + j == A[j] + i$.